



BHABHA UNIVERSITY, BHOPAL

COURSE CURRICULUM FACULTY OF SCIENCE

Postgraduate Programme (Courses effective from Academic Year 2018-19)

SYLLABUS OF COURSES TO BE OFFERED **“Master of Science”** **(ZOOLOGY)**

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Dean faculty of Science, Bhabha University

**Master of Science
(M.Sc.-Zoology)
I-IV Semester**

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M.Sc. – ZOOLOGY

S. No.	Semester	Subject Code	Subject name	Marks
FIRST SEMESTER				
1	SEM-I	MSZ-101	Biosystematics ,Taxonomy And Evolution	100
2		MSZ-102	Structure And Function Of Invertebrates	100
3		MSZ-103	Quantitative Biology, Biodiversity And Wild Life	100
4		MSZ-104	Bio-molecules And Structural Biology	100
5		MSZ- 105	Practical-I	50
6		MSZ-106	Practical-II	50
TOTAL				500
SECOND SEMESTER				
1	SEM-II	MSZ-201	General And Comparative Animal Physiology And Endocrinology	100
2		MSZ-202	Population Ecology And Environmental Physiology	100
3		MSZ-203	Tools And Techniques For Biology	100
4		MSZ-204	Molecular Cell Biology And Genetics	100
5		MSZ-205	Practical	50
6		MSZ-206	Practical	50
TOTAL				500
THIRD SEMESTER				
1	SEM-III	MSZ-301	Comparative Anatomy Of Vertebrates	100
2		MSZ-302	Gamete Biology, Genes, Development & Differentiation	100
3		MSZ-303	Cell Biology	100
4		MSZ-304	Fish Morphology Anatomy & Physiology	100
5		MSZ-305	Practical-I	50
6		MSZ-306	Practical-II	50
TOTAL				500
FOURTH SEMESTER				
1	SEM-IV	MSZ-401	Animal Behaviors And Neurophysiology	100
2		MSZ-402	Eco-Toxicology	100
3		MSZ-403	Taxonomy, Systematic And Ecology Of Fishes	100
4		MSZ -404	Elective-I	100
5		MSZ-405	Practical-I	50
6		MSZ-406	Practical-II	50
Elective-I [optional paper (Choose any one)]				
1		MSZ-404[I]	Cell-Biology (Chromosomes Genes And Genetic Engineering)	100
2		MSZ-404[II]	Pisciculture And Economic Importance Of Fishes	100
3		MSZ-404[III]	Cell-Biology (Neurobiology, Ageing And Immunology)	100
TOTAL				500

M.Sc. Zoology

Semester I

Paper I

Biosystematics, Taxonomy and Evolution

Unit I

· Definition and basic concepts of biosystematics taxonomy and classification.

- History of Classification

Trends in biosystematics : Chemotaxonomy cytotaxonomy and molecular taxonomy

Dimensions of speciation and taxonomic characters.

Species concepts : species category, different species concepts, subspecies and other infra-specific categories.

Theories of biological classification: hierarchy of categories.

Unit II

- Taxonomic Characters – Different kinds.

- Origin of reproductive isolation, biological mechanism of genetic incompatibility.

- Taxonomic procedures: Taxonomic collections , preservation ,cureting, process of identification.

- Taxonomic keys,different types of keys, their merits and demerits.

- International code of Zoological Nomenclature (ICZN): Operative principles, interpretation and application of important rules: Formation of Scientific names of various Taxa.

Unit III

- Taxonomic categories.

- Evaluation of biodiversity indices.

- Evaluation of Shannon – Weiner Index.

- Evaluation of Dominance Index.

- Similarity and Dissimilarity Index.

Unit-IV

- Concepts of evolution and theories of organic evolution.

- Neo Darwinism and population genetics:

- A- Hardy-Weinberg law of genetic equilibrium.

- B – A detailed account of destabilizing forces:
- i- Natural selection
- ii- Mutation
- iii- Genetic Drift
- iv- Migration
- v- Meiotic Drive.
- Trends in Evolution
- Molecular Evolution
- a) Gene evolution
- b) Evolution of gene families
- c) Assessment of molecular variation

Unit – V

- Origin of higher categories
- Phylogenetic – gradualism and punctuated equilibrium.
- Major trends in the origin of higher categories
- Micro and macro evolution.

Molecular population genetics

- Pattern of changes in nucleotide and amino acid sequence.
- Ecological significance of molecular variations (genetic polymorphism)
- Phylogenetic and biological concept of species.
- Patterns and mechanism of reproductive isolation.
- Modes of speciation (allopatry & sympatry)

Origin and Evolution & Economically important microorganisms and animals.

SUGGESTED READING MATERIAL

1. M. Koto-The. Biology of biodiversity-Springer
2. E.O. Wilson-Biodiversity-Academic Press Washington.
3. G.G.-Simpson-Principle of animal taxonomy Oxford IBH Publication company.
4. E-Mayer-Elements of Taxonomy
5. Bastchelet-F-Introduction to mathematics for life scientists Springer Verlag, Berlin.
6. Skoal R.R. and F.J.Rohiff Biometry-Freeman, San-Francisco.
7. Snedcor, G.W. and W.G. Cochran Statistical Methods of affiliated-East-West Press, New Delhi.
8. Murry J.D. Mathematical Biology-Springer, Verlag, Berlin.

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Class - M.Sc.
Subject - Zoology
Paper Title - Paper II STRUCTURE
Semester - I

UNIT –I

1. Origin of metazoa
2. Organization of Coelom
 - A. Acoclomates
 - B. Pscudocoelomates
 - C. Coclomates
3. Locomotion.
 - A. Amoeboid flageller and cillary movement in protozoa
 - B. Hydrostatic movement in Coelenterata
 - C. Annelida and Echinodermata

UNIT –II

A: NUTRITION AND DIGESTION

Patterns of Feeding and digestion in lower metazoa, Mollusea, Echinodermata Filter feeding in polychaeta. B: Respiration

Organs of respiration : Gills, lungs and trachea, respiratory pigments.
Mechanism of respiration.

UNIT – III

EXCRETION

Excretion in lower invertebrates.

Excretion in higher invertebrates.

Mechanism of Osmoregulation.

UNIT – IV

NERVOUS SYSTEM.

- A. Primitive Nervous systems-Coelenterata and Echinodermata.
- B. Advanced nervous system in Annelida, Arthropoda (Crustacea and Insecta) and Mollusa (Cephalopoda)

UNIT – V

A. INVERTEBRATES LARVAL FORMS AND THEIR EVOLUTIONARY SIGNIFICANCE.

- A. Trematoda and Cestoda
- B. Larval forms of Crustacea
- C. Larval forms of Mollusea
- D. Larval forms of Echinodermata.

B. 1. Structure affinities and life history of the following minor noncoelomate Phyla -

- A. Rotifera
- B. Entoprocta

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2. Structure affinities and life history of the following minor Phyla

- A. Phoronida
- B. Ectoprocta

Suggested Reading Material –

1. Hyman, L.H. The invertebrates, Nol. I. protozoa through Ctenophora, McGraw Hill Co., New York
2. Barrington, E.J.W. Invertebrate structure and function. Thomas Nelson and Sons Ltd., London.
3. Jagerstein, G. Evolution of Metazoan life cycle, Academic Press, New York & London.
4. Hyman, L.H. The Invertebrates. Vol. 2. McGraw Hill Co., New York.
5. Hyman, L.H. The Invertebrates. Vol. 8. McGraw Hill Co., New York and London.
6. Barnes, R.D. Invertebrates Zoology, III edition. W.B. Saunders Co. Philadelphia.
7. Russel-Hunter, W.D. A biology of higher invertebrates, the Macmillan Co. Ltd., London.
8. Hyman, L.H. The Invertebrates smaller coelomate groups, Vol. V. McGraw Hill Co., New York.
9. Read, C.P. Animal Parasitism. Parasitism. Prentice Hall Inc., New Jersey.
10. Sedgwick, A.A. Student text book of Zoology. Vol. I, II and III. Central Book Depot, Allahabad.
11. Parker, T.J., Haswell W.A. Text book of Zoology, Macmillan Co., London.

M.Sc. Previous

I Sem III Paper

Quantitative biology, biodiversity and wildlife

Unit – I Quantitative biology

- Basic mathematics for biologists
- matrices and vectors
- Exponential functions
- Differential equations integration
- Periodic functions
- Sprobability distribution properties and probability theory

Unit – II

- Experimental designing and sampling theory
- Completely randomized design and randomized block design
- Analysis of variance
- Co-relation- types of correlation
- Karl persons coefficient correlation
- Regression
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Unit – III Biodiversity

- concept and principal of biodiversity
- causes for the lose of biodiversity
- Biodiversity conservation method
- Medicinal uses of forest plant

Unit – IV Wildlife of India, types of wildlife

- Values of wildlife positive and negative
- Wildlife protection Act
- Conservation of wildlife in India
- Endangered and threatened species

Unit – V Wildlife and conservation

- National Parks and Sanctuaries
- Project Tiger
- Project Gir lion and Crocodile breeding project
- wildlife in M.P. with references to Reptiles Birds and mammals
- Biospheres reserves

Suggested Reading Material:

- Bataschelet. E. Introduction to mathematics for site scientist springer-verlag, berlin
- Jorgenserr, S.E. Fundamental of Ecological modelling E. sevier New York
- Lenderen D. Modelling in behavioral ecology. Chapman & Hall London U.K.
- Sokal, R.R. and F. J. Rohit Biometry Freeman San Francisco
- Snedecor, G.W. and W.G. Cochran, statistical methods, Affiliated East, West Press New Delhi (Indian ed.)
- Muray, J.D. Mathematical Biology, Springer Verlag Berlin
- Pelon, E.C. The interpretation of ecological data : A primer on classification and ordination.
- A. Lewis – Biostatistics
- B.K. Mahajan Methods in Biostatistics

- V.B. Saharia wildlife in India
- S.K. Tiwari wildlife in central India
- J.D. Murrey Mathematical Biology
- Georgs & Wilians Startical method
- R.K. Tondon Biodiversity Texonomy & Ecology
- M.P. Arora An Introduction to prevantology
- P.C. Kotwal Biodiversity and conservation

SEMESTER - I
Paper: IVth Paper
BIOMOLECULES AND STRUCTURAL BIOLOGY

Unit – I

Chemical Foundation of biology

- PH, PK, acids bases, buffers, weak bonds
- Free energy, resonance, isomerisation
- Acid soluble pool of living tissues – aminoacids, monosaccharides, oligosaccharides, nucleotides, peptides.
- Nanoparticles
- Biomaterials

Unit – II

1. Primary, Secondary, tertiary and quaternary structures of proteins, protein folding and denaturation
 2. DNA & RNA: Double helical structure of DNA, Structure of RNA, role of RNA in gene expression
 3. DNA replication, recombination and repair
 4. Functional importance of lipid storage and membrane lipids
 5. Membrane channels and pumps
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1. Basic concepts of metabolism: Coupled and interconnecting reactions of metabolism cellular energy resources and ATP synthesis
 2. Glycolysis and gluconeogenesis
 3. Citric acid cycle
 4. Oxidative phosphorylation : Protein and its regulation
 5. Fatty acid metabolism: Synthesis and degradation of fatty acids
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1. RNA synthesis and splicing
 2. Biosynthesis of amino acids
 3. Biosynthesis of nucleotides
 4. Biosynthesis of membrane lipids and steroids
 5. Protein synthesis

Unit – V

1. Enzymes: Terminologies, classification and basics of enzyme kinetics
2. Mechanism of enzyme catalysis
3. Regulation of enzyme action
4. Concept of free energy and thermodynamic principles in biology
5. Energy rich bonds, compound and biological energy transducers

Suggested Readings:

1. Voet, D. and J.G. Voet. Biochemistry John Wiley & Sons.
2. Freifelder, D. Physical Biochemistry W.H. Freeman & Co.
3. Segal, I.H. Biochemical calculations John Wiley and Sons
4. Creighton, T.E. Protein Structure and Molecular Properties W.H. Freeman & Co.
5. Freifelder, D. Essentials of Molecular Biology
6. Wilson, K. and K.H. Goulding A Biologists Guide to Principles and Techniques of Practical Biochemistry
7. Cooper, T.G. Tools of Biochemistry

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**Class: M.Sc.
SEMESTER - I
Practical : Ist
MAX MARKS-50**

1. Spotting – Classification and identification of various phylum. 10
2. One major dissection of various systems of invertebrates –
Squilla, Prawn, Sepia, Loligo.
10
3. One minor dissection- Grosshopper, Honeybee, Echinus, Starfish, Aplysia.
05
4. Mounting material - permanent balsum mount
05
5. Spottings related with Adaptation. Homologics, Analogics and modification
of mouth parts :
05
6. Viva Voce. 10 And Practical Record-05

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**Class: M.Sc.
SEMESTER - I
Practical : IIInd**

M,M, 50

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| 1. Problem based on Biodiversity and wild life.
Mammals and Fishers group (Spots 5 +5) | 20 |
| 2. Exercise on mean, mode, & Median. | 5 |
| 3. Cell division preparation of slid on Meiosis & Mitosis. | 5 |
| 4. Preparation of different types of chromosomes. | 5 |
| 5. Viva – Voce | 10 |
| 6. Practical Record and collection. | 5 |

Total Marks	<u>50</u>
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Class: M.Sc.

SEMESTER - II

Paper: Ist Paper

GENRAL AND COMPARATIVE ANIMAL PHYSIOLOGY AND ENDOCRONOLOGY

Unit – I

1. Respiratory pigments through different phylogenic groups
2. Transport of oxygen and carbon dioxide in blood and body fluids
3. Regulation of respiration
4. Physiology of impulse transmission through nerves and synapses
5. Autonomic nervous system, neurotransmitters and their physiological functions

Unit – II

1. Patterns of nitrogen excretion in different animal groups
2. Comparative physiology of digestion
3. Osmoregulation in different animal groups
4. Thermoregulation in homeotherms, poikilothermas and hibernation
5. Physiology of pregnancy, placental hormones, pregnancy diagnosis tests, parturition and breast and lactation

Unit – III

1. Comparative study of mechanoreception
2. Comparative study of photoreception
3. Comparative study of phonoreception
4. Comparative study of chemoreception
5. Comparative study of equilibrium reception

Unit – IV

2. Bioluminescence as means of communication among animals
3. Pheromones and other semiochemicals as means of communication among animals
4. Chromatophores and regulation of their function among animals
5. Hormones, their classification and chemical nature
6. Mechanisms of hormone action

Unit –V

1. Phylogeny of endocrine glands (pituitary, pancreas, adrenal, thyroid)
2. Ontogeny of endocrine glands
3. Neuroendocrine system
4. Hormone receptors – signal transduction mechanisms
5. Hormones and reproduction
 - a. Seasonal breeders
 - b. Continuous breeders

**MSc Previous
Subject: Zoology
SEMESTER -II
Paper-I List of Books**

SUGGESTED READING MATERIAL

1. EJW Barrington-General & comparative
Endocrinology-Oxford, Clarendon Press
2. R.H. Williams-Text Book of Endocrinology-W.B. Saunders
3. C.R. Martin- Endocrine Physiology-Oxford University Press.
4. Molecular CellBiology-J. Darnell, H. Lodish and D. Baltimore-
Scientific American Book USA
5. Molecular Biology of the cell-B. Alberts, D-Bray, J.Lewis, M. Raff,
K. Roberts and J.D. Watson, Garland Pub. New York.

M. Sc. Previous

Zoology

Semester II

Paper II

Population Ecology and Environmental physiology

Unit I

1. Populations and their characters.
2. Demography : Life tables, generation time, reproductive value.
3. Population growth: Growth of organisms with non-overlapping generations, stochastic and time lag models of population growth, stable age distribution.
4. Population regulation: Extrinsic and intrinsic mechanisms.

Unit II

1. Adaptations : Levels of adaptations, significance of body size.
2. Aquatic environments : Fresh water, marine, shores and estuarine environments.
3. Eco-physiological adaptations to fresh water environments.
4. Eco-physiological adaptations to marine environments.
5. Eco-physiological adaptations to terrestrial environments.

Unit III

1. Environmental limiting factors.
2. Inter and intra-specific relationship.
3. Predatory- prey relationship, predator dynamics, optimal foraging theory (patch choice, diet choice, prey selectivity, foraging time).
4. Mutualism , evolution of plant pollinator interaction.

Unit IV

1. Conservation management of natural resources .
2. Environmental impact assessment.
3. Sustainable development.

Unit V

1. Concept of homeostasis.
2. Endothermi and physiological mechanism of regulation of the body temperature.
3. Physiological response to oxygen deficient stress.
4. Physiological response to body exercise.
5. Meditation, yoga and their effects.

Suggested Readings:

1. Cherrett, J.M. Ecological Concepts. Blackwell Science Publication, Oxford, U.K.
2. Elseth, B.D. and K.M. Baumgartner, population Biology, Van Nostrand Co., New York.
3. Jorgensen, S.E. Fundamentals of ecological modeling. Elsevier, New York.
4. Krebs, C.J. Ecology. Harper and Row, New York.
5. Krebs, C.J. Ecological Methodology. Harper and Row, New York.
6. Eckert, R. Animal Physiology: Mechanism and Adaptation. W.H. Freeman and Co., New York.
7. Hochachka, P.W. and G.N., Somero. Biochemical adaptation. Princeton, New Jersey.

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SEMESTER - II

Paper: IIIrd Paper

Tools and techniques in Biology

Unit – I

1. Microscopy, principle & applications
 - Light microscope and phase contrast microscope
 - Fluorescence microscope
 - Electron microscope
 - Confocal microscopy
2. General Principle and applications of
 - Colorimeter
 - Spectrophotometer
 - Ultra centrifuge
 - Flame photometer
 - Beer and Lambert's law.
3. Microbiological techniques
 - Media Preparation and sterilization
 - Inoculation and growth monitoring.
 - Microbial assays.
 - Microbial identification (cytological staining methods for bacterial and fungal strains)
 - Use of fermentors

Unit – II

1. Computer aided techniques for data presentation data analysis, statistical techniques.
2. Cryotechniques
 - Cryopreservation of cells, tissues, organs and organisms.
 - Cryosurgery
 - Cryotomy
 - Freeze fracture and freeze drying.
3. Separation techniques. Chromatography, principle type and applications.
 - Electrophoresis, Principles, types and applications PAGE and agarose gel electrophoresis.
 - Organelle separation by centrifugation.

Unit – III

1. Radioisotope and man isotope techniques in biology.
 - a. Sample preparation for radioactive counting
 - b. Autoradiography.
2. Immunological techniques
 - Immunodiffusion (Single & Double)
 - Immuno electrophoresis

3. Techniques immuno detection
 - Immunocyto / histochemistry
 - Immunoblotting, immunodetection, immunofluorescence.
4. Surgical techniques.
 - Organ ablation (eg. Ovariectomy, adrenalectomy)
 - Perfusion techniques
 - Stereotaxy
 - Indwelling catheters
 - Biosensors.

Unit –IV

1. Histological techniques
 - Principles of tissue fixation
 - Microtomy
 - Staining
 - Mounting
 - Histochemistry
2. Cell culture techniques.
 - Design and functioning of tissue culture laboratory
 - Culture media, essential components and Preparation
 - Cell viability testing.

Unit – V

1. Cytological techniques
 - Mitotic and meiotic chromosome preparations from insects and vertebrates.
 - Chromosome banding techniques (G.C.Q. R. banding)
 - Flowcytometry.
 2. Molecular cytological techniques
 - In site hybridization (radio labeled and non-radio labeled methods)
 - Fish
 - Restriction banding
 3. Molecular biology techniques
 - Southern hybridization
 - Northern hybridization -
- DNA Sequencing
- Polymerase chain reaction (PCR)

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**MSc Previous
Subject Zoology
SEMESTER -II
Paper-III Tools & Technique Books**

SUGGESTED READING MATERIAL

1. Introduction to instrumental analysis-Robert Braun-McGraw Hill.
2. A biologist Guide to principles and Techniques of Practical Biochemistry-K, Wilson and K.H. Goulding ElBS Edn.
3. Clark & Swizer. Experimental Biochemistry. Freeman, 2000.
4. Locquin and Langeron. Handbook of Microscopy. Butterwaths, 1983
5. Boyer. Modern Experimental Biochemistry. Benjamin, 1993
6. Freifelder. Physical Biochemistry. Freeman, 1982.
7. Wilson and Wlaker. Practical Biochemistry. Cambridge, 2000.
8. Cooper. The Cell-A Molecular Approach. ASM, 1997
9. John R.W. Masters. Animal Cell culture- A practical approach. IRL Press.
10. Robert Braun. Introduction to instrumental analysis. McGraw Hill

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M.Sc. Previous Zoology

II Sem IV Paper

Topic – Molecular Cell Biology and genetics

- Unit – I Biomembrane
 - Molecular composition arrangement and functional consequences
 - Transport across cell membrane diffusion active transport, pumps, uniports, symports and antiports
 - Micro filaments and microtubules structure and dynamics
 - Cell movements intracellular transport, role of kinesin and dynein
- Unit–II Cell surface receptors
 - Second messenger system
 - Signaling from plasma membrane to nucleus
 - Gap junctions and connexins
 - Integrins
- Unit–III
 - Ca^{++} dependant homophilic cell – cell adhesion
 - Ca^{++} independant homophilic cell – cell adhesion
 - Gap junctions and connexins
 - Genome organization, hierarchy in organization
 - Chromosomal organization of genes and non-coding DNA
- Unit–IV Sex determination
 - Sex determination in Drosophila
 - Sex determination in mammals

- Basic concept of dosage compensation
- Cytogenetic of human chromosomes
- Human genome project (HGP) purpose & Implications
- Unit-V
 - Human gene therapy
 - Prenatal diagnosis & genetic counseling
 - Genetic screening
 - Structural Genomics
 - Functional Genomics
 - Gene libraries
 - Transgenic animals & their applications

Suggested Readings

- J. Darnell, H. Lodish and D. Baltimore molecular cell biology scientific American book. Inc. USA
- B. Alberts D. Bray, J. Lewis, M. Raff, K. Roberts and J.D. Watson. molecular biology of the cell. Garland Publishing Inc. New York.
- John R. W. animal cell culture A practical approach masters. Irl. Press
- Alberts et. al. Essentials cell biology garland publishing Inc. New York 1998
- J.M. Barry molecular biology
- Philip E. Hartman Gene Action
- L.C. Dunn, principles of Genetics
- A.M. Winchester genetics
- Edgar Alterberg Genetics
- L.C. Dunn genetics and the origin of species
- Bengt A. Kihlman actions of chemicals of dividing cells

**Class: M.Sc.
SEMESTER - II
Practical : Ist**

M,M, 50

General & Comarative Physiology and Endocrinology
Population Ecology and Environmental Physiology.

Exercise :

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|--|----|
| 1. Experiment on Hematology Blood group, Total and different counts. | 5 |
| 2. Demonstration of Enzyme Action, and chromatography | 10 |
| 3. Estimation of pH. | 5 |
| 4. Detection of protein carbohydrate and fats. | 5 |
| 5. Endocrinological spots comments on prepared histological slides. | 10 |
| 6. Detection of Nitrogenous products in given samples. | 5 |
| 7. Viva Voce | 5 |
| 8. Practical Records and collection. | 5 |

Total Marks	<u>50</u>
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Class: M.Sc. SEMESTER - II Practical : IInd

M,M, 50

Tools and Techniques for biology. Molecular cell Biology and Genetics

1. Comments upon the structure and application of analytical instruments	10
i. Colorimeter	
ii. Spectrophotometer	
iii. Ultracentrifuge	
iv. ESR and NMR spectrometer	
v. Microtomy	
vi. Chymographic Instruments	
2. Problem and based on genetics	10
3. Estimation techniques based for RNA and DNA	10
4. Estimation of Gene and Genotypic frequencies in light of hardy weinbecey law based on facial traits.	5
5. Demonstration of chromosome polymorphism isozyme polymorphism in some insect population.	5
6. Viva – Voce	5
7. Practical Record	5
Total Marks	<u>50</u>

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Class: M.Sc.
SEMESTER - III
Paper: Ist Paper
Comparative Anatomy of Vertebrates

Unit – I

1. Origin of Chordata : Concept of Protochordata
2. Origin and classification of vertebrates.
3. Vertebrates morphology : Definition, scope and importance.
4. Development, structure and functions of vertebrates integument and its derivatives (glands, scales, feathers and hairs)
5. Respiratory system : Characters of respiratory tissue, external and internal respiration, comparative account of respiratory organs.

Unit – II

1. Evolution of heart
2. Evolution of aortic arches and portal systems
3. Blood circulation in various vertebrates groups
4. Form, function, body size and skeletal elements of the body.
5. Comparative account of jaw suspensorium and vertebral column

Unit – III

1. Comparative account of limbs and girdles.
2. Evolution of urinogenital system in vertebrates.
3. Comparative account of organs of olfaction and taste
4. Comparative anatomy of brain and spinal cord (CNS)
5. Comparative account of peripheral and autonomic nervous system

Unit – IV

1. Comparative account of lateral line system.
2. Comparative account of electroreception.
3. Comparative account of simple receptors.
4. Flight adaptations in vertebrates.
5. Aquatic adaptations in birds and mammals.

Unit – V

1. Origin, evolution general organization and affinities of ostracoderms
2. General organization, specialized, generalized and degenerated characters of cyclostomes.
3. Origin, evolution general organization of early gnathostomes
4. General account of Elasmobranchi, Holocephali, Dipnoi and crossoptergii

Suggested Readings :

1. Carter, G.S. Structure and habit in vertebrate evolution – Sedgwick and Jackson, London.
2. Kingsley, J.S. Outlines of Comparative Autonomy of Vertebrates, Central Book Depot. Allahabad,
3. Kent, C.G. Comparative anatomy of vertebrates
4. Malcom Jollie, Chordata morphology. East – West Pres Pvt. Ltd., New Delhi.
5. Milton I lildergrand. Analysis of vertebrate structure. IV. Ed. John Wiley and Sons Inc., New York.
6. Smith, H.S. Evolution of Chordata structure. Hold Rinchart and Winstoin Inc. New York.
7. Sedgwick, A.A. Students Text Book of Zoology, Vol.II.
8. Walter, H.E. and Sayles, L.D. Biology of vertebrates, MacMillan & Co. New York.
9. Romer, A.S. Vertebrate Body, IIIrd Ed. W.B. Saunders Co., Philadelphia
10. Young J.Z. life of vertebrates. The oxford University Press, London
11. Parker & Haswell to III Rev. by Marshall willians latested Macmillan Co. ltd.
12. Young J.Z. Life of mammals. The Oxford University Press, London
13. Weichert, C.K. and Presch, W. Elements of chordate anatomy, 4th Edn. McGraw Hall Book Co., New York.

M.Sc. Zoology

III Semester

II paper

Gamete Biology, genes development and differentiation

Unit I

1. Comparative account of differentiation of gonads in mammals and invertebrate.
2. Spermatogenesis : Morphological basis in rodents and in any invertebrates. Gamete specific gene expression and genomics
3. Biochemistry of Semen : Semen composition and formation, assessment of sperm function.
4. Fertilization : Prefertilization events Biochemistry of fertilization post fertilization events.

Unit – II

1. Ovarion follicular growth and differentiation : morphology, endocrinology, molecular biology oogenesis and vitellogenesis, ovulation and ovum transport in mammals.
2. Biology of sex determination and sex differentiation a comparative account.
3. Multiple ovulation and embryo transfer technology : in vitro osyte maturation, superevolution.

Unit – III

1. Hormonal regulation of ovulation, pregnancy and parturition.
2. Hormonal regulation of development of mammary gland and lactation.
3. Endocrinology and Physiology of placenta.
4. Collection and cryopreservation of gametes and Embryo.
5. Teratological effects of xenobiotics on gametes.

Unit – IV

1. Cell commitment and differentiation.
2. Germ cell determinants and germ cell migration.
3. Development of gonads.
4. Malanogenesis.

Unit – V

1. Creating new cell types, the basic evolutionary mystery.
2. cell diversification in early embryo, Xenopus, Blastomeres, morphigen gradients, totipotency and pluripotency.
4. Embryonic stem cells, renewal by stem cells, epidermis. Connective tissue cell family
5. Homopoietic stem cells : Blood cells formation, stem cell disorders.

Suggested Readings :

1. Long J.A. Evan H.M. 1922 : the oestrous cycle in the Rat and its associated phenomenon.
2. Nalbandou. A.C. – Reproductive physiology
3. Prakash A.S. 1965-66 Marshall's, Physiology Reproduction (3 Vol.)
4. Gilbert, S.F. Developmenal Biology , Sinauer Associated Inc. Massachulsetts.
5. Ethan Bier, the cold Spring. The cold spring Harbor laboratory Press, New York.
6. Balinsky B.I. Introduction to Embryology sanders, Phliedelpia.
7. Berril N.J. and Karp. G. Development Biology. McGraw Hill New York.
8. Davidson, E.H. Gene Activity During Early Development. Academic Press, New York.

BHABHA UNIVERSITY, BHOPAL

M.Sc. Semester III

Zoology

Paper – III

Cell Biology

Unit - I

1. Principle and applications of spectrophotometer.
2. Cell sorting : Principle and applications of flow cytometer
3. Basic idea of MMR and ESR
4. Principle and application fluorimeter
5. Atomic force microscopy

Unit – II

1. Gel Electrophoresis : 2D- page and isoelectric focusing
2. Stem cells : Embryonic stem cells, culture and application.
3. General idea of X-ray crystallography and its applications
4. Immune techniques : Precipitation, Immunodiffusion, immunoelectrophoresis, ELISA and RIA

Unit – III

1. Methods in protein purification
2. DNA – protein interactions : Electrophoretic mobility shift assay
3. Methods in analysis of gene expression – I : Transformation, transfections and mammalian expression
4. Methods in gene analysis – II : General idea of site directed mutagenesis, Linker scanning mutations analysis Reporter assay.

Unit – IV

Recombinant DNA Technology

1. DNA modifying enzymes of RDT.
2. Properties of RDT
3. Process of RDT
4. Uses of RDT (Recombinant DNA Technology)

General Idea of yeast

1. Two hybrid systems
2. Subtractive hybridization
3. Chromosome walking
4. Chromosome jumping

Unit – V Positional Cloning

1. General idea of RNA se protection assay.
2. Primer extension
3. SI nucleus protection assay
4. Run of transcription and RT PCR

Genome Analysis

1. DNA finger printing
2. RAPD and RFLP

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M.Sc. Zoology

III Semester

IV paper

Fish Morphology Anatomy & Physiology

Unit I

1. Chromatophores, classification, ultra structure and functional significance.
2. Colour changes, Types neural endocrine control mechanism.
3. Respiratory organs – Kinds and physiology of aqueous breathing.
4. Digestive system – Anatomy and Physiology of alimentary canal.

Unit –II

1. Nervous System – Brain its functional organization with ecological bearing.
2. Nervous System – Nerves and their supply.
3. Lateral Line system – structure, modification and significance.
4. Circulatory system in fish, heart venous and arterial system.

Unit –III

1. Neuroendocrine integration in fish.
2. Anatomy and Physiology of pituitary gland,
3. Anatomy and Physiology of thyroid gland.
4. Pineal organ, internal tissue and caudate neurosecretory system.

Unit – IV

1. Environmental and hormonal control of reproduction.
2. Parental care.
3. Early development of a teleost.
4. Osmoregulatory organs mechanisms.

Unit –V

1. Seasonal cycles of gonads in Indian fish.
2. Hormonal and Endocrines control of reproduction in fish.
3. Biochemical composition of fishes.

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M.Sc. Semester III Zoology Practical – I (General)

Time : 5 hour

MM. 50

Comparative anatomy of vertebrates, Gametes Biology of genes, Development and differentiation.

- | | |
|--|----|
| 1. Dissect and display of cranial nerves of Vertebrates | 10 |
| 2. Minor dissection of Herdmania, Accessory respiratory organs perching mechanism. | 5 |
| 3. Prepare a permanent stained slide of given material. | 10 |
| 4. Material and comments upon the spots 1 to 5 including slides and Bones. | 5 |
| 5. Gametes Biology of Fishes, Amphibia, Birds and Mammals. | 5 |
| 6. Identify and comments upon spots on Embryology | 5 |
| 7. Viva Voice | 5 |
| 8. Practical Record. | 5 |

Total Marks

50

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LIST OF PRACTICAL EXERCISES FOR LABORATORY COURSE

M.C. Zoology Semester III

PRACTICAL II

1. Histology and Histochemistry : Microtomy, staining and detection of cell organelles (e.g. Mitochondria Golgi Bodies, lysosomes nucleus and nucleoli)
2. Hisochemical demonstration of lipids, proteins (including enzymes), carbohydrate and nucleic acids (DNA/RNA)
3. Immunocytochemistry : Intracellular localization of specific target molecules by antibody staining.
4. Fluorescence microscopy and immunofluorescence : Application of fluorochromes and flurochrome tagged antibodies in the demonstration of proteins and nucleic acids
5. Get electrophoresis of proteins : Separation of Proteins on polyacrylamide get electrophoresis (PAGE)
6. Gel electrophoresis of nucleic acids (DNA/RNA) Isolation and detection of DNA.RNA on agarose gel.
- 7- Preparation of mitotic chromosomes from ret/ mice bone marrow cells and construct karyotype of G-or C-banded chromosomes
8. Short Terms rat/human blood lymphocyte culture and preparation of mitotic chromosomes for karyotyping.
9. Study of permanent slides and electron micrographs

SCHEME OF PRACTICAL EXAMINATION

1.Histology and histochemistry :	6
a. Demonstration of cellular organelles	
b. Demonstration of biomolecules	
2.Electrophoresis (PAG/agarose)	8
(Demonstration of Bimolecular on gel matrix)	
3.Mitotic chromosome preparations and banding anylysis	8
4.Spotting (permanent slides – 3, electron micrographs – 3)	12
5.Viva Voce	8
6.Practical record.	8

TOTAL MARKS	50
DURATION 6 HOURS	

PROJECT	50
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List of Optional Paper (Special) To be opted to the by the Student in Third Semester and Forth Semester (Paper III and IV). The syllabus of the Optional paper of the will be formed by the respective. University where the Subject are to opted.

Syllabus for same optional papers are is follow.

1. Biology of parasitism and Vertebrate Immune system.
2. Ichthyology (Anatomy and Physiology)
3. Entomology (Applied)
4. Acqua culture and fisheries
5. Endocrinology
6. Wild Life Biology and Conservation
7. Wild Life Conservation and ecotoxicology
8. Microbial Ecology and Biology of Parasitism
9. Limnology
10. Insect Physiology

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Class - M.Sc.
Subject - Zoology
Paper Title - Paper I ANIMAL BEHAVIOUR AND NEUROPHYSIOLOGY
Semester - IV

Max. Marks 50

UNIT I –

1. Introduction :

- Ethology as a branch of biology.
- Animal psychology, classification of behavioral patterns, analysis of behaviour (ethogram)

2. Reflexes and complex behaviour.

3. Preception of the environment: mechanical, electrical, chemical, olfactory, auditory and visual.

4. Evolution and ultimate causation: Inheritance behaviour and relationships.

UNIT II –

1. Neural and hormonal control of behaviour.

2. Genetic and environmental components in the development of behaviour.

3. Motivation: Drive, timing and interaction of drives, physiological basis of motivation, hormones and motivation, aggregation.

4. Communication: Chemical, visual, light and audio, evolution of language (primates).

UNIT III –

1. Ecological aspects of behaviour: Habitat selection, food selection, optimal foraging theory, anti-predator defenses, aggression, homing territoriality, dispersal, host-parasite relations.

2. Biological rhythms: Circadian and circannual rhythms, orientation and navigation, migration of fishes, turtles and birds.

3. Learning and memory: Conditioning, habituation, insight learning, association learning, reasoning.

UNIT IV-

1. Reproductive behaviour. Evolution of sex and reproductive strategies, mating systems, courtship, sexual selection. parental care.
2. Social behaviour. aggregations, schooling in fishes, flocking in birds, herding in mammals, group selection, kin selection, altruism, reciprocal altruism, inclusive fitness, social organization in insects and primates.

UNIT V –

1. Thermoregulation: Homeothermic animals, poikilotherms & Hibernation.
2. Receptor physiology a comparative study –
 - Mechano reception
 - Photo reception
 - Phono reception
 - chemo reception
 - Equilibrium reception
3. Bioluminescence

Suggested Readings -

1. Eibl-Eibesfeldt, I. Ethlogy. The biology of Behaviour. Holt, Rineheart & Winston, New York.
2. Gould, J.L. The mechanism and Evolution of Behaviour.
3. Kerbs, J.R. and N.B. davies : Behaviourable Ecology. Blackwell, Oxford, U.K.
4. Hinde, R.A. Animnal Behaviour : A Synthesis of Ethology and Comparative Psychology. McGraw Hill, New York.
5. Alcock, J. Animal Behaviour : An Evolutionary approach. Sinauer Assoc. Sunderland, Massachsets, USA.
6. Bradbury, J.W. and S.L. Vehrencamp. Principles of Animal Communication. Sinauer Assoc. Sunderland, Massachsets, USA.

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M.Sc. Final
Semester IV
Paper II
Eco-Toxicology

Unit – I

1. General principles of environmental Biology with emphasis on their ecosystems.
2. Abiotic and biotic factor of ecosystems.
3. Communities of the environment, their structure and significance
4. Energy flow in environmentl : Ecological energetics

Unit – II

- 1- Productivity, Production and analysis.
- 2- Recycling and reuse technologic for solid and liquid wastes and their role in environmental conservation
- 3- Remote sensing – basic concept and applications of remote secsing techniques in environmental conservation.
- 4- Environmental indicators and their role in environmental balance

Unit – III

1. Kinds of environmental pollution due to pollutants and their methods
2. Radioactive compounds and their impact on the environment
3. Vehicular exhaust pollution, causes and remedies.
4. Noise pollution

Unit – IV

1. Toxicology basic concepts, Principles and various types of toxicology agents.
2. Toxicity testing principles, hazards, risk and their control methods
3. Food toxicants and their control methods.
4. Public Health Hazards due to environmental disasters.

Unit – V

1. Pesticides, types nature anf their effects on our environment
2. Important heavy, metals and their role in environment.
3. Agrochemical use and misuse , alternatives
4. Occupational Health Hazards and their control.

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M.Sc. Semester IV

Paper II

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List of books-

- | | | |
|----------------------|---|--|
| 1. Arora | : | Fundamentals of environmental |
| 2. Anathakrishnan | : | Bioresources ecology |
| 3. Bottain | : | Environmental studies |
| 4. Bouhey | : | Ecology of populations |
| 5. Clark | : | Elements of ecology |
| 6. Dowdoswell | : | An introduction to animal ecology |
| 7. Goldman | : | Limnology |
| 8. Kormondy | : | Concepts of ecology |
| 9. May | : | Model ecosystems |
| 10. Odum | : | Ecology |
| 11. Perkins | : | Ecology |
| 12. Simmons | : | Ecology of estuaries and costal water |
| 13. Pawlosuske | : | Physico-chemical methods for water |
| 14. South Woods | : | Ecological methods |
| 15. Trivedi and Goel | : | Chemical and biological methods for water
pollution studies |
| 16. Willington | : | Fresh water biology |
| 17. Wetzel | : | Limnology |
| 18. Weleh | : | Limnology Vols. I-II |

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Class: M.Sc.
SEMESTER - IV
Paper: IIIrd Paper
Taxonomy, Systematics and Ecology of Fishes

Unit – I

1. Outline classification of fishes as proposed by Berg.
2. Classification of Elasmobranches
3. Classification of Crossopterygii
4. Classification Actinopterygii

Unit –II

1. Systematic survey of fish with particular reference to inland fishes of M.P.
2. Exotic fishes and their importance.
3. Larvicidal fishes & their importance in Public health.
4. Predatory fishes and their significance in fish culture.

Unit –III

1. Working and maintenance of fish aquarium.
2. Fish nets and gears and methods of fishing.
3. Fish diseases, symptoms and treatment.
4. Common needs of fish Ponds and their control.

Unit – IV

1. Fish parasites and their control.
2. Physics – chemical characteristics of fish pond.
3. Biological characteristics of fish pond.
4. Culturable species of fishes of inland water and basis of their selection.

Unit – V

1. Plankton and their significance in fish culture.
2. Primary productivity of fish ponds and its significance.
3. Ecology of managed fish farm ponds.
4. Ecology of fresh water marshes.

M.Sc. Zoology Semester IV
Paper – IV (Optional - 1)
Cell – Biology
(Chromosomes. Genes and Genetic Engineering)

Unit I

- 1- Molecular Organization of eukaryotic chromosome : Structure of nucleosome particles and higher order compaction of mitotic chromosomes ; chromatin remodelling.
- 2- Specialized chromosomes : structural organization and functional significance of politeness chromosomes
- 3- DNA methylation and DNA Aase – I Hypersensitivity in relational to gene activity and chromatin organization
- 4- Specialized chromosomes II: structural Organization and functional significance of lampbrush chromosomes
- 5- Organization and significance of heterochromatin.

Unit II

1. Structural Organization of eukaryotic genes: Interrupted genes and overlapping genes and their evolution.
2. Gene families: Organization, evolution and significance
3. Transposable genetic elements of prokaryotes and eukaryotes
4. Gene mutations and molecular mechanism of occurrence of mutations Repair Mechanisms

Unit III

1. Organization of eukaryotic transcriptional machinery promoter enhancers transcription factors polymerases activators and repressors.
2. DNA binding domains of transcription apparatus zinc finger steroid receptors homeo domains Helix – loop, Helix and Leucine Zipper.
3. Eukaryotic transcription of eukaryotic transcriptional control
4. Environmental modulation of gene activity (stress response) Stress genes and stress proteins
5. Molecular basis of thalassemias muscular dystrophy cystic fibrosis

Unit IV

1. DNA Rearrangement
2. Amplification during development with special responses to
 - (a) Ciliates
 - (b) Chlorine gene
 - (c) 58 RNA genes
3. Drosophila development I
 - (a) Cleavage
 - (b) Gastrulation
 - (c) Origin of Anterior – Posterior polarity (Maternal effect genes and segmentation)

Class: M.Sc.
SEMESTER - IV
Paper: IVth Paper (Optional -2)
Pisciculture and Economic importance of fishes.

Unit – I

1. Collection of fish seed from natural resources.
2. Dry bundh breeding of camps.
3. Wet bundh breeding of camps.
4. Hypophysation and breeding of Indian major camps.

Unit – II

1. Drugs useful in induced breeding of fish.
2. Types of ponds, required for fish culture farms.
3. Management of hatcheries, nurseries and reany ponds.
4. Management of stocking ponds.

Unit- III

1. Composite fish culture.
2. Prawn culture and pearl industry in India.
3. Fisheries resource of M.P.
4. Riverine fisheries.

Unit – IV

1. Coastal fisheries in India.
2. Offshore and deep sea fisher's in India.
3. Role of fisheries in rural development.
4. Sewage fisheries.

Unit- V

1. Methods of fish Preservation
2. Marketing of fish in India.
3. Economic importance and by products of fishes.
4. Shark liver oil industry in India.
5. Transport of live fish & fish seed.

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M.Sc. Zoology Semester IV Paper – III (Optional - 3) CELL-BIOLOGY (NEUROBIOLOGY, AGEING AND IMMUNOLOGY)

Unit I

1. Voltage gated channels in electrically excitable membrane.
2. C-AMP and calcium as second messengers and their role in cellular regulatory mechanism.
3. Protein phosphorylation and synapsin – I in relation to neuronal regulation
4. Chemical synaptic transmission : Neurotransmitters and role of synaptic vesicles in nerve transmission.

Unit II

1. Neuron : General Organization and functions of nerve fibers
2. Glial cells : Classification and role in nervous system
3. Chromatophores : Types, Structure Composition and functional significance
4. Autonomic neural Regulation of melanophores and color change
5. Intracellular motility : Axonal transport of pigment in chromatophores

Unit III

1. Aging : Theories of aging and current concepts
2. Apoptosis and cell death current concept and significance
3. Age associated neurodegenerative diseases : Alzheimer,s & Parkinson,s diseases
4. Cells and tissues in immune system : General organization and functions
5. Innate and adaptive immunity

Unit IV

1. Antibody – Structure and Function
2. Antibody Diversity
 - (a) Rearrangement in immunoglobulin genes
 - (b) Recombination in immunoglobulin genes
3. Major histocompatibility (MHC) molecules.

Unit V

1. Concept of Innate and acquired Immunity
2. Humoral Immune Responses
3. Cell Mediated Immune Responses
4. Allergy.
5. Auto Immunity
6. Immune Response genes and AIDS

genes

Unit V

4. Drosophila Development II
origin of dorsal ventral polarity
5. Basic Idea of homeotic selector genes and homeotic mutation.
6. Basic Idea of organization of homeoboxes
7. Evolutionary significance of homeoboxes.

Suggested Reading :

- 1- Robertis, De and Robertis Cell and Molecular Biology Lea and Febiger.
2. Watson Hopkins Roberts Steitz and Weiner Molecular Biology of the Gene the Benjamin/Cummings Publishing Company Inc.
3. Bruce Alberts Bray Lewis Raff Roberts Watson Molecular Biology of the Cell. Garland Publishing Inc.
4. Watson Gilman Witkowski Zoller Recombinant DNA Scientific American Books.
5. Karp Gerald Cell Biology.
6. Lewin B. Genes VII
7. King Cell Biology
8. Kaniel L., Hartl , Elizabeth W. Jones. Genetics Principles and Analysis Jones and Bartlett Publisher
9. Lodish Berk Zipursky Matsudaira Baltimore Darnell Molecular Cell Biology W. H. Freeman and Company.
10. Suzuki Griffith Miller Lewontin An Introduction to Genetic Analysis W.H. Freeman and Company
11. Travers, Immunology Current Biology limited.
12. Kuby , Immunology W.H. Freeman and Company.
13. Roitt, Male Snustad Immunology.
14. Gardner Simmons snustad Principles of Genetics John Wiley and Sons Inc.

**Class: M.Sc.
SEMESTER – IV
Practical : Ist (General)**

M,M, 50

1. i. Exercise based on social behaviour : -
Reproductive behaviour
- Learning and Memory
 - ii Experiment based on Animal behavior so as to show – 10
- Taxes
- Reflexes
- Biological Clock
 2. Experiment based on Toxicology testing methods :- 10
- General Test
- Acute toxicity test
- LD 50 test
 3. Pathological techniques in toxicology – autopsy, Histochemical analysis 10
Determination of different Toxic chemicals in samples of soil, water, and air.
 4. Viva 10
 5. Practical Record and Collection 10
- Total Marks

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Class: M.Sc. SEMESTER – IV Practical: IInd (Special)

Time : 5 hours

M,M, 50

- | | | |
|----|---|----|
| 1. | Taxonomical Identification of any 5 fishes | 15 |
| 2. | Identification of Fry and Finger lings. | 5 |
| 3. | Identification of Zoo Plankton | 5 |
| 4. | Spotting on Aquaculture specimen
Mytilus, sepia, Prawn, Shrip Champ, Oysters, Claims, Loligo and
Edible Fishes. | 5 |
| 5. | Identification and comments upon the predatory insects harmful to fish
industry. | 5 |
| 6. | Viva Voce | 10 |
| 7. | Practical Record | 5 |

Project Work

Total Marks